Data Scientist Interview Questions with Answers

Q1. What is Data Science?

Answer: Data Science is the field of extracting insights and knowledge from data using statistics, programming, and machine learning.

Q2. Supervised vs Unsupervised Learning

Answer: Supervised learning uses labeled data; unsupervised learning finds patterns in unlabeled data.

Q3. What is Overfitting?

Answer: Overfitting occurs when a model learns noise in the training data and performs poorly on new data.

Q4. What is Underfitting?

Answer: Underfitting occurs when a model is too simple to capture patterns in the data.

Q5. Bias-Variance Tradeoff

Answer: Bias is error from overly simple assumptions; variance is error from sensitivity to training data.

Q6. What is Cross-Validation?

Answer: A technique for evaluating models by splitting data into multiple train-test folds.

Q7. What is Feature Engineering?

Answer: Creating or transforming variables to improve model performance.

Q8. How do you handle missing values?

Answer: Remove rows, impute with mean/median/mode, or use model-based methods.

Q9. What is Normalization?

Answer: Scaling values to a fixed range, usually 0 to 1.

Q10. What is Standardization?

Answer: Transforming data to have mean 0 and standard deviation 1.

Q11. What is Linear Regression?

Answer: A supervised algorithm used to predict continuous values.

Q12. What is Logistic Regression?

Answer: A classification algorithm used to predict probabilities of classes.

Q13. What is Multicollinearity?

Answer: High correlation among independent variables.

Q14. Regression Metrics

Answer: MAE, MSE, RMSE, and R^2 Score.

Q15. Classification Metrics

Answer: Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

Q16. What is Precision?

Answer: The proportion of predicted positives that are actually positive.

Q17. What is Recall?

Answer: The proportion of actual positives correctly identified.

Q18. What is F1-Score?

Answer: The harmonic mean of precision and recall.

Q19. What is a Confusion Matrix?

Answer: A table showing TP, TN, FP, and FN predictions.

Q20. What is Random Forest?

Answer: An ensemble of decision trees that improves accuracy and reduces overfitting.

Q21. What is XGBoost?

Answer: A high-performance gradient boosting algorithm widely used in competitions and industry.

Q22. What is K-Means?

Answer: A clustering algorithm that groups data into K clusters.

Q23. What is PCA?

Answer: Principal Component Analysis reduces dimensions while retaining important information.

Q24. What is SVM?

Answer: Support Vector Machine finds the best boundary separating classes.

Q25. What is Deep Learning?

Answer: A subset of machine learning using multi-layer neural networks.

Q26. What is NLP?

Answer: Natural Language Processing enables computers to understand human language.

Q27. What is ARIMA?

Answer: A statistical model used for time-series forecasting.

Q28. What is Hypothesis Testing?

Answer: A statistical method for testing assumptions using sample data.

Q29. What is a p-value?

Answer: The probability of observing results assuming the null hypothesis is true.

Q30. What is MLOps?

Answer: Practices for deploying, monitoring, and maintaining machine learning models in production.